Akbar Aman

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Summary

Computer Engineer experienced in designing enterprise AI systems, infrastructure-driven solutions, automation, & production software across industry, research, & client environments.

Education

University of Illinois Chicago, College of Engineering — Expected: May 2027

B.S. in Computer Engineering (Computer & Networked Systems Track), Minor in Mathematics — *GPA: 3.54*

- **Relevant Coursework:** Comp Arch, Networking, Data Structures, Systems Programming, DSP, HDL, VLSI Design
- **Leadership:** Computer Science Teaching Assistant (MATLAB, 2 terms) and Engineering Success Mentor (4 terms).

Experience

Enterprise AI Intern — May 2026 – Present

AHEAD — *Chicago, IL*

- Promoted from a summer internship into a year-round role on AHEAD's Enterprise GPT team, supporting internal R&D, engineering, & deployment while taking technical ownership of an AI skills & agent governance initiative for Glean & Claude. Developed a validated self-service model for creation, discovery, review, & reuse across licensed users.
- Authored the foundational narrative for a proprietary agentic platform, translating its architecture, capabilities, and operating model into a CTO-cleared source of truth for stakeholder approval, internal enablement, & GTM strategy.

AI Safety Evaluator / Red Team Prompt Engineer — Apr 2026 – Aug 2026

LinkedIn — *Remote*

- Red-teamed frontier LLMs through adversarial prompting & structured safety evaluations, eliciting & characterizing failure modes across reasoning, coding, & instruction-following to improve model robustness, reliability, & alignment.

AI-EDGE REU Fellow — Jun 2026 – Jul 2026

NSF AI-EDGE Institute, The Ohio State University — *Remote*

- Selected for the NSF AI-EDGE Summer program at OSU, completing advanced study across LLMs, generative models, AI security, and distributed AI, with applied work in Vision Transformers, agent security, and multi-agent systems.

AI Systems Engineer — Oct 2025 – Mar 2026

Talent Strategy Experts — *Remote*

- Architected & implemented the minimum viable product of a multi-tenant SaaS platform for AI inference, owning end-to-end fullstack, deployment, system documentation, & preparing the system for team-based development & scale. Designed isolation boundaries, role-based access control, and reliability safeguards to support secure org-level data.

Deep Learning/AI Research Intern — Jun 2024 – Aug 2024

UIC Department of Electrical & Computer Engineering — *Chicago, IL*

- Led a team of five to design and prototype *Vision Mamba*, a custom neural network achieving +30% accuracy over baselines while optimizing for low-latency, high-throughput inference on constrained edge devices, and completed specialized advanced HW/SW co-design training using hls4ml to synthesize PyTorch models into FPGA bitstreams.

Technical Projects

Governed Platform for Support (GPS) | *Python, FastAPI, PostgreSQL/pgvector, RAG, LLM Orchestration* [GitHub]

- **Independently authored** a provider-agnostic governed AI platform for secure, grounded automation across org support stacks. Designed a modular agent runtime with vector retrieval, durable case/run state, typed decision contracts, human-approval gates, deterministic tool authorization, and reconstructable end-to-end audit trails.

Full-Stack MLS Property Platform (IDX Exchange Internship) | *Python, PHP, MySQL, REST API* [Live]

- **Led a SWE team of 6**; created a real-time property search platform integrating live MLS data via RESTful APIs utilizing OAuth 2.0; engineered automated cron-job synchronization and optimized MySQL schemas to manage 1,000+ active listings on a Linux VPS environment while ensuring efficient data normalization and low-latency.

Bare-Metal Embedded Safety System | *C, ARM Assembly, CCS, Cortex-M4F* [GitHub]

- Engineered a **dual-mode** bare-metal system on ARM Cortex-M4F utilizing Timer Capture ISRs, ADC sequencers, and Memory Mapped I/O-based PWM drivers to achieve precision ranging and 0.1°F resolution, while implementing a non-blocking architecture via Wait-For-Interrupt (WFI) to minimize power consumption during idle execution cycles.

Technical Skills

Credentials: AWS: Agentic AI; Anthropic: Claude API/MCP/Bedrock; Google: Deploy Multi-Agent Architectures

Languages: Python, C++, C, JavaScript, TypeScript, Java, SQL, PHP, Bash, ARM/MIPS Assembly, SystemVerilog

Technologies: AWS, GCP, Docker, Linux, CI/CD, REST APIs, React, Node.js, PyTorch, Terraform, Vivado, FPGA